

upctl-compose 用户手册

本地部署 & AI Agent 自动处理指南

基于 AI Agent 的全自动开发平台

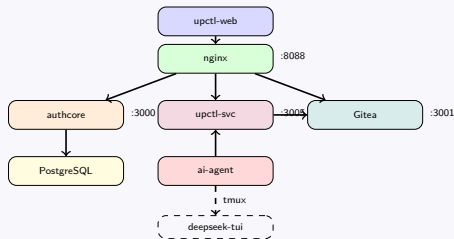
馨翼

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系统架构总览



服务	端口	作用
nginx	8088	反向代理，路由到各后端
authcore	3000	身份认证，JWT 签发
upctl-svc	3005	工单 CRUD API
upctl-web	(nginx 内)	Vue 3 前端
ai-agent	—	轮询 Gitea + tmux 托管 deepseek-tui
Gitea	3001	代码托管 + 工单 + CI
PostgreSQL	5432	数据库

环境要求

软件要求

- Docker 24.0+, Compose 2.24+
- Git

端口占用

5432 PostgreSQL	3000 authcore
3001 Gitea	3005 upctl-svc
8088 nginx	2222 Gitea SSH

推荐配置

- CPU 4 核 +、内存 8 GB +
- 需访问 GitHub + DeepSeek API

注意： Docker Desktop 需配置 no_proxy（已内置）

克隆与启动

1. 克隆

```
git clone https://github.com/  
alchemy-studio/upctl-compose.git  
cd upctl-compose
```

2. 启动

```
docker compose up -d  
docker compose ps
```

启动日志

```
authcore: Running migration...  
[authcore] Starting htyuc...  
数据库连接池初始化 OK  
upctl-svc: starting on port 3005  
ai-agent: Setting up workspace...  
Starting tmux session...
```

初始化 Gitea

等待 Gitea

```
until curl -sf localhost:3001/api/v1/version  
do sleep 3; done
```

初始化

```
docker compose exec ai-agent \  
python3 /app/setup-gitea.py
```

创建内容

组织 weli + 仓库 tickets

标签: approved(1) / in_progress(2) / blocked(3)

管理界面

<http://localhost:3001>

用户 ai-bot / ai-bot-dev-pass

AI 代理配置

.env 文件

```
DEEPSEEK_API_KEY=sk-your-key  
DEEPSEEK_API_BASE=https://api.deepseek.com  
DEFAULT_MODEL=deepseek-v4-flash
```

重启

```
docker compose up -d ai-agent
```

验证部署

```
curl -sf localhost:8088/ | head -c 200  
curl -sf localhost:3005/ # upctl-svc  
curl -sf localhost:3000/ # AuthCore
```


登录页面

访问 `http://localhost:8088`

- 开发模式显示用户名密码登录表单
- 使用 demo / demo123 登录
- AuthCore 验证后直接签发 JWT
- 登录后跳转工单列表

认证流程： 前端 → nginx → authcore 验证密码
→ JWT



工单列表 & 详情

列表页：标签页切换 / 卡片显示标题 + 状态

详情页：标题 + 编号 / 状态标签 / 评论区

AI 处理结果格式：

Processing result

{AI 总结}

--- *Processed by DeepSeek Agent*

The screenshot displays a web application for managing tickets. At the top, there's a header with the title '工单列表' (Ticket List) and a '添加' (Add) button. Below the header, there are three tabs: '待处理' (Pending), '已关闭' (Closed), and '全部' (All). The main content area shows a list of tickets. Two tickets are visible:

- Ticket #24: 数据库连接池优化 (Database connection pool optimization), status '待处理' (Pending), created at 05-09 11:35.
- Ticket #23: 登录页面样式调整 (Login page style adjustment), status '待处理' (Pending), created at 05-09 11:35.

Below the ticket list, there's a code editor showing a Python script for handling orders. The code includes imports for 'os', 'python', 'import csv', 'from django.http import HttpResponseRedirect', 'from django.core.urlresolvers import login_required', and 'from models import WorkOrder'. It defines a function 'login_required' and a view function 'def export_workorders_csv(request)' that handles CSV export requests. The view function checks the request status, filters orders, and returns a CSV response.

```
#!/usr/bin/env python
# coding: utf-8

import os
import csv
from django.http import HttpResponseRedirect
from django.core.urlresolvers import login_required
from models import WorkOrder

@login_required
def export_workorders_csv(request):
    status = request.GET.get('status', '')
    orders = WorkOrder.objects.all()
    if status:
        orders = orders.filter(status=status)

    response = HttpResponseRedirect('')
    response['Content-Disposition'] = 'attachment; filename="workorders.csv"'

    writer = csv.writer(response)
    writer.writerow(['ID', 'Title', 'Status', 'Assigned To', 'Created At', 'Updated At'])
    for order in orders:
        writer.writerow([order.id, order.title, order.get_status_display(),
            order.assigned_to.username if order.assigned_to else '',
            order.created_at, order.updated_at])
```

Gitea 后台

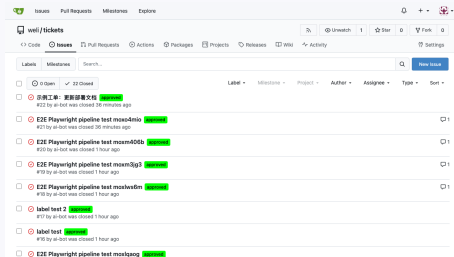
地址 `http://localhost:3001`

用户 `ai-bot` / `ai-bot-dev-pass`

仓库 `weli/tickets`

标签说明：

- `approved` — AI 自动处理
- `in_progress` — 处理中
- `blocked` — 阻塞



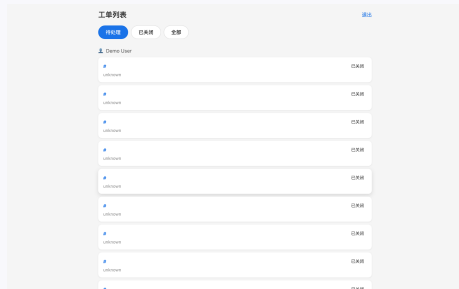
AI Agent 自动处理

系统部署完成后，AI Agent 自动开始轮询：

- ① 在 Gitea 创建工单并打 approved 标签
- ② ai-agent 每 5 分钟轮询发现新工单
- ③ 自动调用 DeepSeek API 处理
- ④ 评论结果并关闭工单

也可手动启动即时处理：

```
$ docker compose exec ai-agent \
  python3 /app/poll_worker.py --once
```



ai-agent 自动轮询日志：

```
Processing issue #185
DeepSeek response received
close #185: 200 -- completed
```

各组件日志解读

authcore 启动

```
Running migration 2022-02-14-124946
Running migration 2026-05-02-082153
[authcore] Starting htyuc...
全局数据库连接池初始化 OK
```

upctl-svc

```
[upctl-svc] starting on port 3005
[upctl-svc] listening on 0.0.0.0:3005
```

ai-agent 轮询

```
poll_worker started - interval=300s
No approved issues found.
No work. Sleeping 300s...
```

ai-agent 处理

```
Processing issue #185: 添加 dependbot
add_label #185: 200
DeepSeek response received (594 chars)
add_comment #185: 200
close #185: 200 - completed
```

tmux 运行时 & 容器状态

tmux 调试

```
# 进入 session
docker compose exec ai-agent
  tmux attach -t deepseek-agent
# 查看输出
docker compose exec ai-agent
  tmux capture-pane -t deepseek-agent -p
```

docker compose ps

NAME	STATUS
ai-agent	Up (healthy)
authcore	Up
gitea	Up
postgres	Up (healthy)
upctl-svc	Up
upctl-web	Up

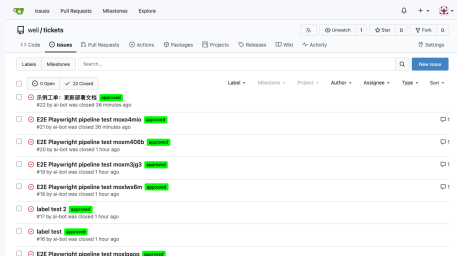
使用 tmux send-keys 注入命令

可在不进入交互模式的情况下向 agent 发送命令

Gitea Issue 自动处理流程

本系统使用 Gitea + ai-agent 实现工单自动处理，无需外部 CI：

- 1 创建工单，标记 approved
- 2 ai-agent 每 5min 轮询 Gitea API
- 3 发现 approved 工单 → 加 in_progress
- 4 tmux send-keys 注入工单内容到 deepseek-tui 对话
- 5 deepseek-tui 与 api.deepseek.com 交互生成回复
- 6 tmux capture-pane 读取回复内容
- 7 评论结果到工单
- 8 移除 in_progress + 关闭工单



ai-agent 工作日志

启动阶段

```
Setting up workspace...
Starting tmux session...
Starting poll worker...
poll_worker started - interval=300s
```

空闲状态

```
No approved issues found.
No work. Sleeping 300s...
```

处理工单

```
Processing issue #185: 添加 dependbot
add_label #185: 200
DeepSeek response received (594 chars)
add_comment #185: 200
close #185: 200 - completed
```

关键技术： Python 轮询 | tmux send-keys 注入 | deepseek-tui CLI | Gitea API

deepseek-tui 验证

tmux 终端

```
root@ai-agent:/app/workspace# deepseek-tui --version
deepseek-tui (npm wrapper) v0.8.22
binary version: v0.8.22
repo: git+https://github.com/Hmbown/DeepSeek-TUI.git
```

注意： 日志已脱敏，无密文泄露

实际运行日志

```
docker compose logs ai-agent --tail 5
[ai-agent] Setting up workspace...
[ai-agent] Starting tmux session...
[ai-agent] Starting poll worker...
[2026-05-09] poll_worker started -- interval=300s
[2026-05-09] No work. Sleeping 300s...
```

常见问题

问题	原因	解决
POOL_SIZE must be set	authcore 缺变量	加 POOL_SIZE: "10"
JWT_KEY not set	upctl-svc 缺变量	加 JWT_KEY 环境变量
Gitea API 404	路径错误	用 /api/v1/ 根路径
前端白屏/JS 404	资源未构建	docker compose build upctl-web
ai-agent 未连 DeepSeek	缺 API Key	配置后重启
容器间连接失败	Docker Desktop 代理	设置 no_proxy (已内置)
Playwright 401	JWT 格式	ts 无 Z 后缀
remove_label 404	标签 ID	approved=1, in_progress=2

总结

upctl-compose AI Agent 驱动的全自动开发平台

核心能力



6 容器一键部署



DeepSeek 自动处理



Playwright 全链路测试

```
git clone https://github.com/alchemy-studio/upctl-compose.git && cd  
upctl-compose && docker compose up -d
```

Thank You

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<https://github.com/alchemy-studio/upctl-compose>